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DATA320 Writing Assignment 3

1. The first video delves into an introductory walkthrough for probability and statistics so that one can have a foundational understanding of the field to succeed in the future. From the collection, summary, and interpretation of information to the use of data to answer questions about the natural world, the video tells the viewer about how populations and samples interact with one another prior to the breakdown of real statistical topics. Examples of how data can help people draw insights and inform decisions were a major point of contention within my first viewing of the video. Upon watching it again, I saw recurring topics from what we discussed last week about datasets and the ways in which they are processed and collected before being loaded into our usable environment. This made the procedures simple and intuitive for practical use, as sample viewings were more reliable than trying to gauge the entire population directly in most, if not all, cases.
2. The second video looks at basic data visualizations alongside variables and the different ways in which they're used to categorize the narratives that we see. It starts off by explaining the major differences between categorical and numerical (quantitative) variables, and then moves into discussing how each of these types is used in various statistical analyses. Discrete and continuous parameter variables are placed within bar charts, pie charts, histograms, stem plots, time plots, and more to show datasets in a new

light that wasn't previously understandable through a look at the data alone. Seeing these examples again after class helped drill the concepts for practicality.

3. This video introduces basic statistical concepts surrounding variance and description. It explains what the mean, median, and mode mean in their proper context so that viewers can understand what central tendency means within datasets. It also speaks on how these measures show up differently depending on the type of statistical distribution you might be analyzing. Although we were introduced to these concepts in class many times throughout the years, this video showed them under a new lens that reinforced the importance of changing how you use these measures when you have outliers in your data.
4. This fourth video focused on the effects of transforming data based on the spread and center, and looked at this variability alongside measures like standard deviation. It provided examples of relative frequency situations and gave word problems that asked to understand the values of things like the mean within the proper context. The art of transforming data to draw meaningful insights beyond what you can simply examine from the numbers within the observations and variables is important. This class and this video combined to help me deeply understand that from a concrete perspective. It's also especially important for me as an aspiring data scientist, as many libraries and frameworks require you to understand what you're doing on a basic level before you begin performing deeper analysis and tests.
5. This fifth video went into the concept of outliers and how they affect the general spread and center tendency or variance on a basic level. In introducing things like range, mean, median, and standard deviation again, one can realize that spread-out data points affect these measures in strange ways, depending on the skew and concentration of points

within the density distributions. Realizing how variability interacts with your specific set of data is important because numerical data can manifest different trends in what seems to be the same way on the surface (Simpson's Paradox). Only through aggregations and visualizations can these differences become quite visible and accounted for. This topic was faintly touched on in class, but the depth to which it could be applied was only revealed after I watched this video.

6. The sixth video introduces summary, boxplots, and outliers with undertones of probability and how it relates to the general subsequent analysis. The function of boxplots allows one to see how certain particular outcomes are, so that reliable insights into the future can be underscored valuably. From the minimum, quartiles, median, and the maximum, the distribution of each point in the plot is displayed so that one can understand how the data spreads throughout accordingly. Then, from Q3 to Q1, there is an inter-quartile-range (IQR) that makes up the actual box and houses the median so that the user of the plot can identify where most of these observations should fall. Anything outside of this and along the line by a certain degree above Q3 ($Q3 + 1.5 * IQR$) and below Q1 ($Q1 - 1.5 * IQR$) is considered an outlier. These concepts were briefly touched on in class, but a review of the feasible nature of these elements was helpful for my basic understanding.
7. This video briefly touched on percentiles and described them as the percentage of data values that fall at or below another specified data value. For instance, the 50th percentile meant 50% of the data points are either at this point or below this point. Quartiles corresponded to percentiles at each point they represented. If some data point was considered to mark the 3rd quartile, then that also meant that it marked the 75th percentile

for that distribution. In real-world contexts, I've seen this in national test distributions in which I would score in a percentile that placed me above a certain set of kids throughout the nation. Anyone below that percentile scored worse than I, and anyone above it performed slightly better. The video used the same context and applied it to math exam scores. In class, we briefly mentioned this when going over visualizations and how they interact with one another in real-world examples. However, seeing a concept like this again and applying it to actual scenarios that I barely even thought about before was extremely worthwhile.

8. The last video delved into symmetry and skewness and how they play a part in the insights we can draw from our data. Depending on the skew of our distribution, the mean might be greater than or less than the median. For instance, if the distribution showed left skew (longer left tail of outliers), then the average of the entire plot would be brought down, and the bins of the histogram overshadowing the density plot would hold a mean that is less than the median value, since that typically marks the middle (Resists the changes brought about by the lower end values). By this same token, if the distribution showed right skew (longer right tail of outliers), then the average of the entire plot would rise up, and the bins of the histogram overshadowing the density plot would hold a mean that is greater than the median value since that typically marks the middle (Resists the changes brought about by the higher end values). This video is directly connected to the last few classes that we've had, and was particularly helpful because it looked at multiple different visualizations and showed how they were affected after applying the same symmetry and skewness statistical concepts that were initially just applied to the histogram plots.

9. The presentation style of the videos was similar to that of a class on statistical concepts, with examples used to make them more understandable for the general audience who interacts with them. On the most basic level, they both aid in introducing the concepts of foundational data science and how they interact with statistics on a fundamental level. However, I didn't find the videos that much more helpful than simply being exposed to the concepts in class and practicing with them on my own outside of class through more academic resources and avenues. I understand the sentiment and the means by which they can be helpful, but a lot of information being wrestled with is too elementary for the level of math that we are already expected to be bringing into the course (And therefore needlessly time-consuming). Class itself is much more structured and detailed than those short-form videos could ever hope to be. They are far too condensed, and look at these concepts on the most basic level instead of combining and utilizing them for a greater and more difficult purpose (Not good for deeper understanding or practice).
10. The author has created these videos to help aspiring students of statistics learn these concepts on the most basic and easy level. They're good as they simplify things that are typically complex. They definitely help the general population learn material that is typically not as accessible for most audiences without a math background. In building up this platform on YouTube as a resource for students, it can serve more than one functional purpose, depending on what it helps. For example, right now it's being used as the main teaching resource, but in the future, one might use it to verify the sanctity of what they're learning prior to consulting other resources.
11. The second and third videos felt relatively long for the little amount of material that they covered, as they focused on basic concepts without defining clearer learning objectives.

With an introduction to fairly simple concepts, there was definitely time for the creator of the videos to explain things in a quicker and less convoluted way. Videos that spoke on percentiles and the effect of outliers were way too short, since these concepts can be explained at a much deeper level than what the creator was willing to get into. There are many real-world scenarios and formulas that call for a proper understanding of percentiles, quartiles, and percentages in general.

12. A major group that benefits from this style of short-form video is high-school students who are loosely interested in applying statistics in their future academic careers on the most basic level. From there, they can learn what they desire at a deeper level before eventually entering the workforce with their knowledge, since these videos explain things rather clearly as an introduction. In addition, those in the workforce who need a simple refresher of these concepts to do the job at hand will find this breakdown of the fundamentals to be essential for their occupation. Many employees in the field of data analytics, business, and sciences will find these methodologies useful for whatever they might find themselves encountering. Furthermore, populations that are generally interested in what statistics has to offer will find these videos great for their learning experiences, as it's often a field of importance that is overlooked by many. Without a basic introduction into the major topics that statistics may offer, people can't hope to truly understand how they themselves are represented in datasets throughout the world (Political, healthcare, employment, financial, etc.).